

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

PROJECT REPORT

Submitted by **SUBASREE G**

25ITR144

JAVAASSIGNMENT

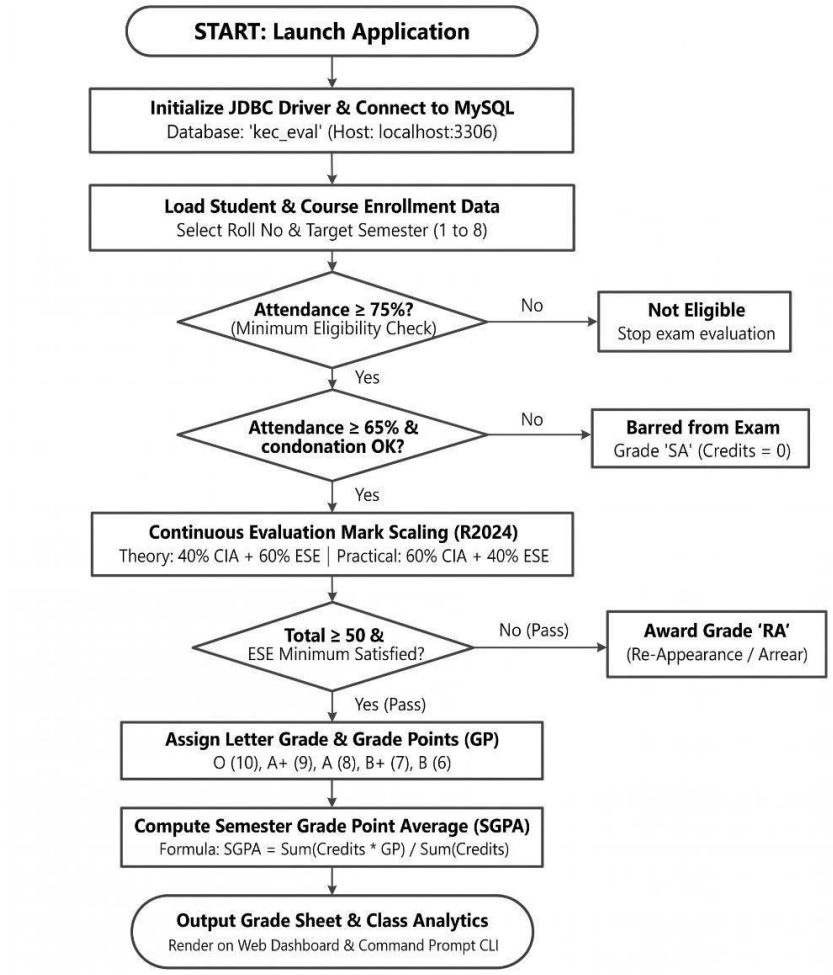
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_eval database running on the local MySQL server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

- Generate student grade sheets, attendance shortage reports, and department-wise pass percentage reports. Display the results through the interactive web dashboard and the command prompt console**.**

DATABASE SCHEMA:

1. STUDENTS TABLE:

```
mysql> DESC students;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| roll_no         | varchar(20)  | NO   | PRI | NULL    |       |
| student_name    | varchar(100) | NO   |     | NULL    |       |
| department      | varchar(50)  | NO   |     | NULL    |       |
| current_semester | int         | NO   |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
4 rows in set (0.01 sec)
```

2. COURSES TABLE:

```
mysql> DESC courses;
+-------------+-----------------------------------------+------+-----+---------+-------+
| Field       | Type                                    | Null | Key | Default | Extra |
+-------------+-----------------------------------------+------+-----+---------+-------+
| course_code | varchar(20)                             | NO   | PRI | NULL    |       |
| course_title | varchar(100)                           | NO   |     | NULL    |       |
| credits     | int                                     | NO   |     | NULL    |       |
| course_type | enum('THEORY','PRACTICAL','EMBEDDED')   | NO   |     | NULL    |       |
+-------------+-----------------------------------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

3. ENROLLMENTS TABLE:

```
mysql> DESC enrollments;
+--------------------+-------------+------+-----+---------+----------------+
| Field              | Type        | Null | Key | Default | Extra          |
+--------------------+-------------+------+-----+---------+----------------+
| enrollment_id      | int         | NO   | PRI | NULL    | auto_increment |
| roll_no            | varchar(20) | NO   | MUL | NULL    |                |
| course_code        | varchar(20) | NO   | MUL | NULL    |                |
| semester           | int         | NO   |     | NULL    |                |
| classes_held       | int         | YES  |     | 0       |                |
| classes_attended   | int         | YES  |     | 0       |                |
| medical_condonation | tinyint(1) | YES  |     | 0       |                |
+--------------------+-------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

4. ASSESSMENT MARKS TABLE:

```
mysql> DESC assessment_marks;
+----------------+--------------+------+-----+---------+-------+
| Field          | Type         | Null | Key | Default | Extra |
+----------------+--------------+------+-----+---------+-------+
| enrollment_id  | int          | NO   | PRI | NULL    |       |
| cia1           | decimal(5,2) | YES  |     | NULL    |       |
| cia2           | decimal(5,2) | YES  |     | NULL    |       |
| assignments    | decimal(5,2) | YES  |     | NULL    |       |
| quizzes        | decimal(5,2) | YES  |     | NULL    |       |
| lab_evaluation | decimal(5,2) | YES  |     | NULL    |       |
| ese_marks      | decimal(5,2) | YES  |     | NULL    |       |
| ese_absent     | tinyint(1)   | YES  |     | 0       |       |
+----------------+--------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

SOURCE CODE:

MAINAPP.JAVA:

```
import java.sql.*;

import java.util.Scanner;



public class MainApp {

    private static final Scanner SC = new Scanner(System.in);

    private static final StudentDAO STUDENTS = new StudentDAO();

    private static final AcademicDAO ACADEMICS = new AcademicDAO();



    public static void main(String[] args) {

        try (Connection ignored = DBConnection.getConnection()) {

            System.out.println("MySQL connection successful.");

        } catch (SQLException ex) {

            System.out.println("Database connection failed: " + ex.getMessage());

            System.out.println("Check MySQL service, schema, Connector/J JAR, and DBConnection credentials.");

            return;

        }
```

```
while (true) {
    System.out.println("\n========== KEC STUDENT ACADEMIC SYSTEM ==========");
    System.out.println("1. Add Student");
    System.out.println("2. Add Course");
    System.out.println("3. Enroll Student in Course");
    System.out.println("4. Add/Update Assessment Marks");
    System.out.println("5. Update Attendance");
    System.out.println("6. View Student Semester Mark Sheet");
    System.out.println("7. Attendance Defaulter Report");
    System.out.println("8. Department Result Summary");
    System.out.println("0. Exit");
    int choice = readInt("Enter choice: ");
    try {
        switch (choice) {
            case 1: addStudent(); break;
            case 2: addCourse(); break;
            case 3: enroll(); break;
            case 4: marks(); break;
            case 5: attendance(); break;
            case 6: ACADEMICS.printMarkSheet(readLine("Enter Roll Number: ").toUpperCase()); break;
            case 7: ACADEMICS.printAttendanceDefaulters(); break;
            case 8: ACADEMICS.printDepartmentSummary(); break;
            case 0: System.out.println("Thank you for using KEC Student Academic System."); return;
            default: System.out.println("Invalid choice. Please select from the menu.");
        }
```

```
} catch (Exception ex) {
            System.out.println("Operation failed: " + ex.getMessage());
        }
    }
}

private static void addStudent() throws SQLException {
    String roll=readLine("Roll Number (e.g. 24ITR001): ").toUpperCase();
    String name=readLine("Student Name: ");
    String dept=readLine("Department (CSE/ECE/MECH/IT): ").toUpperCase();
    int sem=readInt("Current Semester (1-8): ");
    STUDENTS.addStudent(roll,name,dept,sem);
    System.out.println("Student added successfully.");
}
private static void addCourse() throws SQLException {
    String code=readLine("Course Code (e.g. 24ITR31): ").toUpperCase();
    String title=readLine("Course Title: ");
    double credits=readDouble("Credits: ");
    String type=readLine("Course Type (THEORY/PRACTICAL/EMBEDDED): ").toUpperCase();
    ACADEMICS.addCourse(code,title,credits,type);
    System.out.println("Course added successfully.");
}
private static void enroll() throws SQLException {
    String roll=readLine("Roll Number: ").toUpperCase();
    String code=readLine("Course Code: ").toUpperCase();
    int sem=readInt("Semester: ");
```

```
 long id=ACADEMICS.enroll(roll,code,sem);
        System.out.println("Enrollment successful. Enrollment ID = " + id + " (use this ID for marks/attendance).");
    }
    private static void marks() throws SQLException {
        long id=readLong("Enrollment ID: ");
        Double c1=readOptionalMark("CIA Test 1 (0-100, blank to keep/zero): ");
        Double c2=readOptionalMark("CIA Test 2 (0-100, blank to keep/zero): ");
        Double aql=readOptionalMark("Assignment/Quiz/Lab (0-100, blank to keep/zero): ");
        String absentInput=readLine("Absent in ESE? (Y/N): ").trim();
        boolean absent=absentInput.equalsIgnoreCase("Y");
        Double ese=absent ? null : readOptionalMark("ESE Marks (0-100, blank to keep/zero): ");
        ACADEMICS.saveMarks(id,c1,c2,aql,ese,absent);
        System.out.println("Assessment marks saved successfully.");
    }
    private static void attendance() throws SQLException {
        long id=readLong("Enrollment ID: ");
        int held=readInt("Total classes held: ");
        int attended=readInt("Classes attended: ");
        double pct=held==0 ? 0 : attended*100.0/held;
        boolean condonation=false;
        if(pct>=65 && pct<75) condonation=readLine("Medical condonation approved? (Y/N):
").equalsIgnoreCase("Y");
        ACADEMICS.updateAttendance(id,held,attended,condonation);
        System.out.printf("Attendance updated: %.2f%%%n",pct);
    }
    private static String readLine(String prompt) {
        System.out.print(prompt);
```

```
        return SC.nextLine().trim();
    }
    private static int readInt(String prompt) {
while(true) { try { return Integer.parseInt(readLine(prompt)); } catch(NumberFormatException ex) {
System.out.println("Enter a valid whole number."); } }
    }
    private static long readLong(String prompt) {
     while(true) { try { return Long.parseLong(readLine(prompt)); } catch(NumberFormatException ex) {
System.out.println("Enter a valid ID number."); } }
    }
    private static double readDouble(String prompt) {
        while(true) { try { return Double.parseDouble(readLine(prompt)); } catch(NumberFormatException ex) {
System.out.println("Enter a valid number."); } }
    }
    private static Double readOptionalMark(String prompt) {
        while(true) {
            String value=readLine(prompt);
            if(value.isEmpty()) return 0.0;
            try {
                double mark=Double.parseDouble(value);
                if(mark>=0 && mark<=100) return mark;
                System.out.println("Marks must be between 0 and 100.");
            } catch(NumberFormatException ex) { System.out.println("Enter a valid mark from 0 to 100."); }
        }
    }
}
```

DBCONNECTION.JAVA:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
public final class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/kec_academic_db";
    private static final String USER = "root";
    private static final String PASSWORD = "Suba@1324";
    private DBConnection() { }
    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

OUTPUT SCREENSHOTS:

MAIN MENU:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 1
```

1. ADDING STUDENT:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 1
Roll Number (e.g. 24ITR001): 25ITR124
Student Name: SAMRUKA KK
Department (CSE/ECE/MECH/IT): IT
Current Semester (1-8): 3
Student added successfully.
```

2. ADD COURSE:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 2
Course Code (e.g. 24ITR31): 25ITC04
Course Title: EVS
Credits: 1
Course Type (THEORY/PRACTICAL/EMBEDDED): THEORY
Course added successfully.
```

3. STUDENT ENROLLMENT:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 3
Roll Number: 25ITR144
Course Code: 24ITC45
Semester: 3
Enrollment successful. Enrollment ID = 8 (use this ID for marks/attendance).
```

4. UPDATE MARKS:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 4
Enrollment ID: 2
CIA Test 1 (0-100, blank to keep/zero): 85
CIA Test 2 (0-100, blank to keep/zero): 80
Assignment/Quiz/Lab (0-100, blank to keep/zero): 90
Absent in ESE? (Y/N): N
ESE Marks (0-100, blank to keep/zero): 94
Assessment marks saved successfully.
```

5. UPDATE ATTENDANCE:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 5
Enrollment ID: 2
Total classes held: 60
Classes attended: 54
Attendance updated: 90.00%
```

6. STUDENT MARKSHEET:

```
========== KEC STUDENT SEMESTER MARK SHEET ==========

Student: Subasree G | 25ITR144 | Dept: IT | Sem: 3
Course: 24ITC45 - DESIGN LAB (PRACTICAL), Credits: 2.0
CIA1: 0.00, CIA2: 0.00, Assignment/Quiz/Lab: 0.00, CIA Avg: 0.00
Internal weighted: 0.00, ESE raw: 0.00, ESE weighted: 0.00, Total: 0.00
Attendance: 0.00% | Grade: SA | Status: BARRED - SHORTAGE OF ATTENDANCE

Student: Subasree G | 25ITR144 | Dept: IT | Sem: 3
Course: 25ITC31 - Full Stack (THEORY), Credits: 3.0
CIA1: 85.00, CIA2: 80.00, Assignment/Quiz/Lab: 90.00, CIA Avg: 85.00
Internal weighted: 34.00, ESE raw: 94.00, ESE weighted: 56.40, Total: 90.40
Attendance: 90.00% | Grade: A+ | Status: PASS

SGPA: 5.40
Equivalent Percentage (SGPA x 10): 54.00%
================================================
```

7. ATTENDANCE DEFAULT REPORT:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 7

--- ATTENDANCE DEFAULTERS (<65%) : SA ---
25ITR144 | Subasree G | IT | Course 25ITC31 | 50.00% | SA
```

8.DEPARTMENT RESULT SUMMARY:

```
========= KEC STUDENT ACADEMIC SYSTEM =========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice:
Enter a valid whole number.
Enter choice: 8

--- DEPARTMENT RESULT SUMMARY ---
ECE | Enrolled course results: 1 | Pass count (approx): 1 | Fail count (approx): 0
IT | Enrolled course results: 3 | Pass count (approx): 0 | Fail count (approx): 3
MECH | Enrolled course results: 1 | Pass count (approx): 0 | Fail count (approx): 1
Note: Class average SGPA is not included in this compact summary; calculate per-student SGPA from the mark-sheet output.
```

DATABASE:

STUDENTS DATABASE:

```
mysql> SELECT * FROM students;
+----------+--------------+------------+------------------+
| roll_no  | student_name | department | current_semester |
+----------+--------------+------------+------------------+
| 23EER122 | Hari S       | ECE        |                7 |
| 23ITR156 | Arun k       | IT         |                7 |
| 23MER164 | Trishna K    | MECH       |                4 |
| 24ECR136 | Fathima I    | ECE        |                5 |
| 24MER178 | Rithika S    | MECH       |                5 |
| 25CSR132 | Kani J       | CSE        |                3 |
| 25CSR144 | Kumaran G    | CSE        |                3 |
| 25ITR121 | Sahana S     | IT         |                3 |
| 25ITR124 | SAMRUKA KK   | IT         |                3 |
| 25ITR144 | Subasree G   | IT         |                3 |
+----------+--------------+------------+------------------+
10 rows in set (0.00 sec)
```

COURSES DATABASE:

```
mysql> SELECT * FROM courses;
+-------------+--------------+---------+-------------+
| course_code | course_title | credits | course_type |
+-------------+--------------+---------+-------------+
| 22ITC32     | PC           |       3 | PRACTICAL   |
| 23ITP06     | DLD          |       2 | PRACTICAL   |
| 24ITC31     | ITC          |       4 | THEORY      |
| 24ITC45     | DESIGN LAB   |       2 | PRACTICAL   |
| 24ITP34     | JAVA         |       4 | THEORY      |
| 25ITC04     | EVS          |       1 | THEORY      |
| 25ITC31     | Full Stack   |       3 | THEORY      |
| 25ITP07     | PLDS         |       3 | EMBEDDED    |
+-------------+--------------+---------+-------------+
8 rows in set (0.00 sec)
```

ENROLLMENTS DATABASE:

```
mysql> SELECT * FROM enrollments;
+--------------+----------+------------+----------+-------------+-----------------+--------------------+
| enrollment_id | roll_no  | course_code | semester | classes_held | classes_attended | medical_condonation |
+--------------+----------+------------+----------+-------------+-----------------+--------------------+
|            2 | 25ITR144 | 25ITC31    |        3 |          60 |              30 |                  0 |
|            3 | 24ECR136 | 24ITP34    |        5 |           0 |               0 |                  0 |
|            5 | 24MER178 | 24ITP34    |        5 |          25 |              24 |                  0 |
|            6 | 25ITR124 | 24ITC45    |        3 |           0 |               0 |                  0 |
|            8 | 25ITR144 | 24ITC45    |        3 |           0 |               0 |                  0 |
+--------------+----------+------------+----------+-------------+-----------------+--------------------+
5 rows in set (0.00 sec)
```

ASSESSMENT MARKS DATABASE:

```
mysql> SELECT * FROM assessment_marks;
+---------------+-------+-------+-------------+---------+----------------+-----------+------------+
| enrollment_id | cia1  | cia2  | assignments | quizzes | lab_evaluation | ese_marks | ese_absent |
+---------------+-------+-------+-------------+---------+----------------+-----------+------------+
|             2 | 85.00 | 80.00 |       90.00 |    NULL |           NULL |     94.00 |          0 |
|             3 | 60.00 | 85.00 |       90.00 |    NULL |           NULL |     90.00 |          0 |
|             5 | 84.00 | 97.00 |       85.00 |    NULL |           NULL |      NULL |          1 |
|             6 |  NULL |  NULL |        NULL |    NULL |           NULL |      NULL |          0 |
|             8 |  NULL |  NULL |        NULL |    NULL |           NULL |      NULL |          0 |
+---------------+-------+-------+-------------+---------+----------------+-----------+------------+
5 rows in set (0.00 sec)
```

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

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GITHUB:

LINK:	https://github.com/subasreeg2007-creator/jdbc_java